

2024 Ph H2 Q5

Section: The Expanding Universe
Topic: Redshift, Wavelength, Olber’s Paradox

(a)

Determine the wavelength of the light represented by the students.

Given:

- Students = wavefronts
- Emitted every 3.5 s
- Students walk at 1.5 m/s

Wavelength = distance between wavefronts
= speed × time interval

$\lambda = v \times t = 1.5 \text{ m/s} \times 3.5 \text{ s} = 5.25 \text{ m}$

Answer:
5.3 m (2 sig. fig. to match given data)

Marks: 3

- 1 mark: correct method
- 1 mark: correct substitution
- 1 mark: correct answer

(b)

State if the ‘star’ is moving towards or away from the observer.
You must justify your answer.

In the diagram, the wavefronts are more spread out than before.

This means:

- The wavelength has increased
- The frequency has decreased

This is the definition of redshift → occurs when the source is moving away.

Answer:
Away from the observer

Justification: The wavefronts are further apart than before, indicating redshift.

Marks: 2

- 1 mark: correct direction
- 1 mark: justified using redshift/wavelength spacing

(c)(i)

Explain how Olber’s paradox supports the concept of the expanding Universe.

Olber’s paradox:
If the Universe were infinite, static, and filled with stars, the night sky should be bright in every direction.

But we see a dark night sky.
This supports the idea that:

- The Universe is expanding and
- Has a finite age, so light from very distant stars hasn’t reached us yet.

Marks: 2

- 1 mark: statement of paradox
- 1 mark: link to expanding Universe and finite age

(c)(ii)

State one other piece of evidence that supports the concept of the expanding Universe.

Acceptable answers include:

- Cosmic Microwave Background Radiation (CMBR)
- Abundance of light elements (Hydrogen, Helium)
- Observed redshift of galaxies

Answer:
CMBR

Mark: 1

Revision Tips:

- Redshift = light’s wavelength increases → star is moving away
- The faster the star moves, the greater the redshift
- Olber’s paradox shows the night sky is dark → Universe cannot be infinite and unchanging
- CMBR is leftover heat from the Big Bang, detected as microwave background radiation